

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

- First Line Contains Integer n – Size of array
- Next n lines Contains n numbers – Elements of an array
- Last Line Contains Integer x – Value for x

Output Format

- First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int findFloor(int arr[], int low, int high, int x) {
3     if (low > high) return -1;
4     if (x >= arr[high]) return arr[high];
5     int mid = low + (high - low) / 2;
6     if (arr[mid] == x || (mid > 0 && arr[mid - 1] <= x && x < arr[mid]))
7         return arr[mid] == x ? arr[mid] : arr[mid - 1];
8     if (x < arr[mid])
9         return findFloor(arr, low, mid - 1, x);
10    return findFloor(arr, mid + 1, high, x);
11 }
12 int main() {
13     int n, x;
14     if (scanf("%d", &n) != 1) return 0;
15     int arr[n];
16     for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
17     if (scanf("%d", &x) != 1) return 0;
18     printf("%d\n", findFloor(arr, 0, n - 1, x));
19     return 0;
20 }
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓
✓	7 3 5 7 9 11 13 15 10	9	9	✓

Passed all tests! ✓

Correct